Farmers Market Database

Basic SQL Selection & Sorting Interview Guide

Focus: SELECT, ORDER BY, *

Introduction

This document contains 15 interview-level SQL questions based on the **Farmers Market** database. **Constraint:** These questions strictly test your ability to query data using only `SELECT`, `FROM`, `ORDER BY`, and the wildcard `*`.

The Challenge Questions

Basic Projection (SELECT & *)

1. **Question 1:** *Full Inventory Dump:* The manager needs a complete backup of the product category list. Write a query to select absolutely all data columns and rows from the `product_category` table.
2. **Question 2:** *Vendor Contact List:* We need to generate a simple calling list. Retrieve only the `vendor_name`, `vendor_owner_first_name`, and `vendor_owner_last_name` from the `vendor` table.
3. **Question 3:** *Customer Zip Codes:* To map our demographic reach, select just the `customer_zip` column from the `customer` table.
4. **Question 4:** *Product Sizes:* Select the `product_name` and its corresponding `product_size` from the `product` table.
5. **Question 5:** *Booth Identification:* Select the `booth_number` and `booth_type` from the `booth` table.

Sorting Data (ORDER BY)

6. **Question 6:** *Alphabetical Vendors:* List all columns for every vendor, but ensure the output is sorted alphabetically by the `vendor_name` from A to Z.
7. **Question 7:** *Reverse Customer Sort:* Retrieve all customers (all columns), but sort them by their `customer_last_name` in descending order (Z to A).
8. **Question 8:** *Chronological Markets:* Select the `market_date` and `market_day` from the `market_date_info` table, ordered from the earliest date to the latest date.
9. **Question 9:** *Highest Price First:* From the `vendor_inventory` table, select the `product_id` and `original_price`, sorting the results so the most expensive items appear at the very top.
10. **Question 10:** *Product Category Order:* Select all columns from the `product` table, ordered by the `product_category_id` in ascending order.

Multi-Column Sorting & Complex Projection

11. **Question 11:** *Regional Sorting:* Retrieve the `customer_first_name`, `customer_last_name`, and `customer_zip` from the `customer` table. Sort the results first by `customer_zip` (ascending), and then by `customer_last_name` (ascending) for customers within the same zip code.

12. **Question 12:** *Booth Pricing Tiers:* Select the `booth_number`, `booth_price_level`, and `booth_description` Sort the list primarily by `booth_price_level` (A, B, C...) and secondarily by `booth_number` (lowest to highest).

13. **Question 13:** *Inventory Management:* Select the `vendor_id`, `product_id`, and `quantity` from the `vendor_inventory` table. Sort by `quantity` descending (largest stock first), and break ties using `vendor_id` ascending.

14. **Question 14:** *Vendor Type Listing:* Select the `vendor_type` and `vendor_name` from the `vendor` table. Sort the list alphabetically by `vendor_type` first, then by `vendor_name` descending.

15. **Question 15:** *Date and Time Sort:* Select the `market_date`, `transaction_time`, and `customer_id` from the `customer_purchases` table. Order the output by `market_date` descending, and then by `transaction_time` ascending (earliest transactions for that day first).

Answer Key (SQL Workbench Syntax)

```sql
SELECT * FROM product_category;
```

Answer 1: Full Inventory Dump

```sql
SELECT
    vendor_name,
    vendor_owner_first_name,
    vendor_owner_last_name
FROM vendor;
```

Answer 2: Vendor Contact List

```sql
SELECT customer_zip
FROM customer;
```

Answer 3: Customer Zip Codes

```sql
SELECT product_name, product_size
FROM product;
```

Answer 4: Product Sizes

```sql
SELECT booth_number, booth_type
FROM booth;
```

Answer 5: Booth Identification

```sql
SELECT * FROM vendor
ORDER BY vendor_name ASC;
```

Answer 6: Alphabetical Vendors

```sql
SELECT * FROM customer
ORDER BY customer_last_name DESC;
```

Answer 7: Reverse Customer Sort

```sql
SELECT market_date, market_day
FROM market_date_info
ORDER BY market_date ASC;
```

Answer 8: Chronological Markets

```sql
SELECT product_id, original_price
FROM vendor_inventory
ORDER BY original_price DESC;
```

Answer 9: Highest Price First

```sql
SELECT * FROM product
ORDER BY product_category_id ASC;
```

Answer 10: Product Category Order

```sql
SELECT
    customer_first_name,
    customer_last_name,
    customer_zip
FROM customer
ORDER BY customer_zip ASC, customer_last_name ASC;
```

Answer 11: Regional Sorting

```sql
SELECT
    booth_number,
    booth_price_level,
    booth_description
FROM booth
ORDER BY booth_price_level ASC, booth_number ASC;
```

Answer 12: Booth Pricing Tiers

```sql
SELECT
    vendor_id,
    product_id,
    quantity
FROM vendor_inventory
ORDER BY quantity DESC, vendor_id ASC;
```

Answer 13: Inventory Management

```sql
SELECT vendor_type, vendor_name
FROM vendor
ORDER BY vendor_type ASC, vendor_name DESC;
```

Answer 14: Vendor Type Listing

```sql
SELECT
    market_date,
    transaction_time,
    customer_id
FROM customer_purchases
ORDER BY market_date DESC, transaction_time ASC;
```

Answer 15: Date and Time Sort